

## GIMLI AP, MB, Canada

WMO: 717480

Lat: 50.6394N

Lon: 97.0460W

Elev: 755

StdP: 14.30

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -27.7      | -22.7 | -35.7                         | 0.7 | -27.4 | -30.5 | 1.0 | -22.4 | 28.9                  | 10.2 | 25.9 | 7.9  | 5.4                   | 280  | 0.682 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.7                   | 84.0            | 69.5 | 81.0 | 67.5 | 78.4 | 65.8 | 72.3                | 80.4 | 70.1 | 78.0 | 68.1 | 75.6 | 12.5                 | 190  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 69.6                            | 111.9 | 77.2 | 67.2 | 103.0 | 75.0 | 65.2 | 95.8 | 72.7 | 36.6          | 80.5 | 34.6 | 78.1 | 32.9 | 75.7 | 84.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |    |    |
| 25.3              | 22.3 | 19.8 | -35.4                      | 90.4 | 5.2                | 3.8 | -39.1                                              | 93.2 | -42.2      | 95.4 | -45.1      | 97.6 | -48.8      | 100.3 |    |    |
|                   |      |      | -35.4                      | 76.4 | 5.4                | 2.9 | -39.3                                              | 78.5 | -42.5      | 80.1 | -45.5      | 81.7 | -49.4      | 83.8  |    |    |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 36.2   | 1.9   | 5.0   | 19.7  | 36.7  | 49.7  | 61.4  | 66.0  | 64.0  | 55.0  | 41.2  | 23.9  | 7.9   |
|                                                                   | DBStd       | 25.19  | 14.87 | 13.86 | 14.15 | 10.93 | 8.79  | 6.97  | 5.53  | 6.02  | 7.93  | 8.48  | 11.95 | 14.19 |
|                                                                   | HDD50       | 6645   | 1492  | 1259  | 941   | 411   | 115   | 5     | 0     | 0     | 39    | 296   | 782   | 1305  |
|                                                                   | HDD65       | 10715  | 1957  | 1679  | 1406  | 848   | 478   | 147   | 55    | 93    | 312   | 738   | 1232  | 1770  |
|                                                                   | CDD50       | 1607   | 0     | 0     | 1     | 13    | 106   | 346   | 496   | 433   | 189   | 23    | 0     | 0     |
|                                                                   | CDD65       | 201    | 0     | 0     | 0     | 0     | 4     | 38    | 86    | 61    | 12    | 0     | 0     | 0     |
|                                                                   | CDH74       | 1849   | 0     | 0     | 0     | 3     | 89    | 355   | 732   | 557   | 110   | 3     | 0     | 0     |
|                                                                   | CDH80       | 373    | 0     | 0     | 0     | 0     | 16    | 75    | 143   | 124   | 15    | 0     | 0     | 0     |
|                                                                   | WSAvg       | 9.2    | 9.3   | 9.2   | 9.5   | 9.7   | 10.0  | 8.9   | 8.0   | 8.2   | 8.9   | 9.9   | 9.9   | 9.0   |
|                                                                   | PrecAvg     | 20.8   | 0.8   | 0.6   | 1.1   | 1.4   | 2.4   | 3.4   | 3.1   | 2.7   | 2.0   | 1.5   | 1.0   | 1.0   |
| Precipitation                                                     | PrecMax     | 30.2   | 2.0   | 1.4   | 3.0   | 2.2   | 4.3   | 7.2   | 5.8   | 4.6   | 4.5   | 3.1   | 3.0   | 2.6   |
|                                                                   | PrecMin     | 13.3   | 0.2   | 0.2   | 0.1   | 0.7   | 1.0   | 1.5   | 1.6   | 0.8   | 0.7   | 0.2   | 0.3   | 0.2   |
|                                                                   | PrecStd     | 4.4    | 0.6   | 0.4   | 0.8   | 0.5   | 1.1   | 1.7   | 1.1   | 1.1   | 1.1   | 0.9   | 0.6   | 0.5   |
|                                                                   | PrecStd     | 4.4    | 0.6   | 0.4   | 0.8   | 0.5   | 1.1   | 1.7   | 1.1   | 1.1   | 1.1   | 0.9   | 0.6   | 0.5   |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 37.6  | 38.6  | 55.8  | 72.2  | 82.0  | 87.1  | 87.7  | 87.9  | 82.4  | 70.3  | 54.6  | 38.1  |
|                                                                   |             | MCWB   | 34.6  | 35.1  | 45.8  | 54.8  | 60.5  | 71.3  | 73.5  | 71.5  | 66.2  | 58.1  | 46.5  | 34.1  |
|                                                                   | 2%          | DB     | 32.6  | 34.7  | 46.7  | 64.7  | 75.9  | 81.4  | 83.7  | 83.4  | 77.1  | 64.0  | 47.6  | 33.2  |
|                                                                   |             | MCWB   | 30.8  | 32.9  | 40.7  | 49.7  | 57.7  | 66.5  | 70.7  | 69.2  | 63.8  | 54.0  | 42.2  | 31.2  |
|                                                                   | 5%          | DB     | 27.8  | 31.8  | 41.1  | 58.6  | 70.7  | 77.9  | 80.9  | 80.0  | 72.8  | 59.0  | 43.0  | 30.5  |
|                                                                   |             | MCWB   | 26.4  | 30.4  | 37.2  | 47.1  | 55.7  | 64.3  | 68.6  | 67.7  | 61.6  | 51.5  | 39.1  | 29.1  |
|                                                                   | 10%         | DB     | 23.6  | 26.2  | 37.3  | 53.4  | 65.5  | 74.7  | 78.2  | 76.7  | 68.7  | 54.5  | 38.7  | 27.5  |
|                                                                   |             | MCWB   | 22.4  | 24.8  | 34.4  | 44.2  | 53.7  | 63.1  | 67.1  | 65.8  | 60.0  | 48.7  | 35.9  | 26.2  |
|                                                                   | 0.4%        | WB     | 34.5  | 35.6  | 48.5  | 56.0  | 64.7  | 74.6  | 76.4  | 74.0  | 69.7  | 59.7  | 47.8  | 34.6  |
|                                                                   |             | MCDB   | 36.4  | 37.5  | 53.7  | 68.4  | 75.9  | 84.8  | 85.1  | 82.6  | 77.5  | 68.6  | 52.5  | 37.3  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 30.5  | 33.3  | 41.3  | 51.2  | 60.6  | 69.7  | 73.2  | 71.3  | 66.1  | 55.8  | 43.2  | 31.6  |
|                                                                   |             | MCDB   | 32.3  | 34.7  | 46.0  | 63.2  | 70.3  | 77.5  | 80.8  | 80.0  | 72.5  | 61.8  | 47.2  | 33.1  |
|                                                                   | 5%          | WB     | 26.3  | 30.1  | 37.4  | 48.1  | 57.6  | 66.8  | 70.8  | 69.3  | 63.7  | 52.4  | 39.4  | 29.0  |
|                                                                   |             | MCDB   | 27.7  | 32.0  | 41.0  | 56.9  | 66.5  | 74.4  | 78.3  | 77.5  | 70.1  | 58.0  | 42.9  | 30.4  |
|                                                                   | 10%         | WB     | 22.3  | 24.9  | 34.6  | 45.1  | 55.1  | 64.6  | 68.6  | 67.4  | 61.1  | 48.9  | 36.1  | 26.1  |
|                                                                   |             | MCDB   | 23.5  | 26.4  | 37.0  | 53.2  | 64.4  | 71.8  | 75.9  | 74.7  | 67.7  | 54.3  | 38.9  | 27.4  |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 17.4  | 19.0  | 18.5  | 19.7  | 20.7  | 18.9  | 19.7  | 20.5  | 19.4  | 15.7  | 14.0  | 15.7  |
|                                                                   |             | MCDBR  | 17.8  | 16.3  | 18.3  | 28.2  | 29.7  | 24.2  | 23.1  | 24.8  | 25.7  | 23.4  | 19.1  | 16.3  |
|                                                                   |             | MCWBR  | 16.6  | 14.8  | 13.0  | 16.7  | 15.3  | 12.4  | 12.8  | 13.2  | 14.2  | 14.7  | 13.9  | 14.8  |
|                                                                   | 5% WB       | MCDBR  | 17.9  | 16.5  | 17.1  | 26.1  | 23.2  | 20.1  | 20.8  | 22.4  | 21.1  | 21.5  | 17.6  | 16.2  |
|                                                                   |             | MCWBR  | 16.7  | 14.9  | 12.6  | 16.3  | 13.8  | 11.8  | 12.8  | 12.9  | 13.2  | 14.3  | 13.2  | 14.7  |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.263 | 0.296 | 0.306 | 0.354 | 0.368 | 0.384 | 0.393 | 0.377 | 0.346 | 0.300 | 0.271 | 0.247 |
|                                                                   | taud        |        | 2.223 | 2.136 | 2.278 | 2.334 | 2.371 | 2.355 | 2.357 | 2.405 | 2.479 | 2.559 | 2.423 | 2.289 |
|                                                                   | Ebn at Noon |        | 243   | 261   | 285   | 282   | 282   | 277   | 272   | 272   | 269   | 266   | 244   | 234   |
|                                                                   | Edh at Noon |        | 22    | 32    | 34    | 36    | 36    | 37    | 37    | 33    | 28    | 21    | 18    | 18    |
|                                                                   | RadAvg      |        | 387   | 758   | 1165  | 1597  | 1729  | 1835  | 1913  | 1589  | 1125  | 652   | 349   | 280   |
| All-Sky Solar Radiation                                           | RadStd      |        | 38    | 55    | 87    | 108   | 147   | 144   | 88    | 112   | 101   | 84    | 49    | 28    |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | -1.26 | -1.73 | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (19 neighbors) | N/A   | N/A     | N/A    | N/A     | -1.28 | -1.78 | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page